20 Solve algebraically the simultaneous equations

$$y = 2x^2 - 3x - 10$$
$$2x - y = -2$$

$2x - y = -2 \rightarrow y = 2x + 2$

$2x^2 - 3x - 10 = 2x + 2$

1 mark
Eliminate y by substitution.

$2x^2 - 5x - 12 = 0$

1 mark
Rearrange to a quadratic=0.

	x	-4
$2x$	$2x^2$	$-8x$
3	$3x$	-12

$(2x + 3)(x - 4) = 0$

1 mark
Factorise to solve.

$x = -1.5$ or $x = 4$

1 mark
Both x roots.

If $x = -1.5$, $y = 2(-1.5) + 2 = -1$

If $x = 4$, $y = 2(4) + 2 = 10$

$(x, y) = (-1.5, -1)$ or $(4, 10)$

1 mark
Correctly match each x and y.

(Total for Question 20 is 5 marks)